

LABORATORY OBSERVATION/RECORD

Course:	Full Stack Web Development
Course Code:	23CM4121
Experiment No.:	2
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1. TITLE

Design Webpage Using HTML5 & CSS3

2. AIM

To design a visually styled profile card webpage using HTML5 for structure and CSS3 for styling, including custom fonts, gradients, flexbox layout, hover effects, and responsive design.

3. OBJECTIVE

- To structure webpage content using HTML5 elements.
- To apply CSS3 styling using internal stylesheets and CSS custom properties (variables).
- To use Flexbox for arranging and aligning page elements.
- To style typography using imported Google Fonts.
- To create visual depth using box-shadow, gradients, and border-radius.
- To apply hover and focus-visible states for interactive elements.
- To make the webpage responsive using CSS media queries.
- To respect accessibility preferences using the prefers-reduced-motion media feature.

4. THEORY

HTML5 provides the structural elements of a webpage, while CSS3 controls its visual presentation. Together they allow a plain document to be transformed into a polished, interactive interface. In this experiment, a single HTML file contains both the semantic markup and an internal `<style>` block that defines the complete visual design of a personal profile card.

CSS Custom Properties

The `:root` selector defines CSS variables such as `--paper`, `--ink`, `--gold`, and `--teal`. These variables store reusable colour and sizing values, which are referenced throughout the stylesheet using the `var()` function, making the design easy to maintain and re-theme.

Flexbox Layout

The body element uses `display: flex` to centre the profile card both vertically and horizontally on the page. The `.socials` container also uses flexbox to arrange the Email and LinkedIn links evenly in a row.

Typography and Web Fonts

Google Fonts (Fraunces, Inter, and IBM Plex Mono) are imported using `<link>` tags, giving the heading, body text, and metadata labels each a distinct typographic voice.

Visual Depth and Decoration

Layered box-shadow values simulate a soft drop shadow beneath the card, a radial-gradient background adds subtle lighting, and repeating-linear-gradient patterns create a faint paper-grain texture across the page.

Interactivity and Accessibility

The `:hover` and `:focus-visible` pseudo-classes change the background, colour, and position of the social links when a user points at or tabs to them. A `prefers-reduced-motion` media query disables the hover transition for users who have requested reduced motion at the operating-system level, and a `max-width` media query reflows the layout for small screens.

The basic flow of the application is:

HTML Structure (card, text, links) → Internal CSS3 Styling (variables, flexbox, gradients, fonts) → Browser Rendering → Styled, Responsive Profile Card

5. ALGORITHM / PROCEDURE

- 1 Create a new HTML5 document with the DOCTYPE declaration and a descriptive page title.
- 2 Add a viewport meta tag to support responsive rendering on mobile devices.
- 3 Add `preconnect` and `stylesheet <link>` tags to import the Fraunces, Inter, and IBM Plex Mono Google Fonts.
- 4 Add an internal `<style>` block inside the `<head>` for all CSS rules.
- 5 Define CSS custom properties inside `:root` for the colour palette and card width.
- 6 Apply a universal box-sizing: `border-box` rule to all elements.
- 7 Style the body using flexbox to centre content, and apply a radial-gradient background.
- 8 Add a `body::before` pseudo-element with repeating-linear-gradient to create a subtle grid texture.
- 9 Create a `.card` container with padding, border-radius, border, and layered box-shadow values.
- 10 Add a circular `.stamp` badge, positioned absolutely, with a dashed inner ring using `::after`.
- 11 Style the `.eyebrow` label, the `h1` heading, and the italic `.role` subtitle using the imported fonts.
- 12 Style the `.bio` paragraph text and a dashed `.divider` rule beneath it.
- 13 Create a `.meta-row` using flexbox with `justify-content: space-between` for the location and availability details.
- 14 Create a `.socials` flex container holding the Email and LinkedIn links.
- 15 Style each social link as a bordered, rounded button containing an inline SVG icon and label text.
- 16 Add `:hover` and `:focus-visible` states that change the background, text colour, and vertical position of each link.
- 17 Add a `max-width` media query so the card becomes full width and the social buttons wrap on small screens.
- 18 Add a `prefers-reduced-motion` media query to remove the hover transition when requested.
- 19 Build the `<body>` markup: the `.card` div containing the stamp, eyebrow, heading, role, bio, divider, meta row, and social links.

20 Save the file as index.html and open it in a browser to verify the styled output.

21 Test the hover effects, keyboard focus states, and responsive behaviour at different screen widths.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Govindamma – Profile Card</title>

<link rel="preconnect" href="https://fonts.googleapis.com">
<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
<link href="https://fonts.googleapis.com/css2?family=Fraunces:ital,opsz,wght@0,9..144,300;0,9..144,500;0,9..144,600;1,9..144,500&family=Inter:wght@400;500;600&family=IBM+Plex+Mono:wght@400;500&display=swap"
rel="stylesheet">

<style>
:root {
  --paper: #E8E3D6;
  --paper-dark: #DED8C7;
  --ink: #201C17;
  --ink-soft: #4A443B;
  --gold: #A9793F;
  --teal: #2C4A52;
  --line: #C6BEA9;
  --card-w: 380px;
}

* { box-sizing: border-box; }

body {
  margin: 0;
  min-height: 100vh;
  display: flex;
  align-items: center;
  justify-content: center;
  background:
    radial-gradient(circle at 20% 15%, rgba(255,255,255,0.5), transparent 45%),
    var(--paper-dark);
  font-family: 'Inter', sans-serif;
  padding: 40px 20px;
}

body::before {
  content: "";
  position: fixed;
  inset: 0;
  pointer-events: none;
  opacity: 0.5;
  background-image:
    repeating-linear-gradient(0deg, rgba(0,0,0,0.015) 0px, transparent 1px, transparent 2px),
    repeating-linear-gradient(90deg, rgba(0,0,0,0.015) 0px, transparent 1px, transparent 2px);
}

.card {
  position: relative;
  width: var(--card-w);
  background: var(--paper);
  border: 1px solid var(--line);
  border-radius: 14px;
  padding: 46px 34px 30px;
  box-shadow:
    0 1px 0 rgba(255,255,255,0.6) inset,
    0 30px 60px -20px rgba(32,28,23,0.35),
    0 10px 20px -12px rgba(32,28,23,0.25);
}

.stamp {
  position: absolute;
  top: -22px;
  right: 28px;
  width: 58px;
  height: 58px;
  border-radius: 50%;
  background: var(--teal);
  color: var(--paper);
  display: flex;
```

```

    align-items: center;
    justify-content: center;
    font-family: 'Fraunces', serif;
    font-size: 17px;
    font-weight: 500;
    letter-spacing: 0.5px;
    transform: rotate(-9deg);
    box-shadow: 0 8px 16px -6px rgba(44,74,82,0.55);
    border: 2px solid var(--paper);
}
.stamp::after {
    content: "";
    position: absolute;
    inset: 6px;
    border-radius: 50%;
    border: 1px dashed rgba(232,227,214,0.55);
}

.eyebrow {
    font-family: 'IBM Plex Mono', monospace;
    font-size: 11px;
    letter-spacing: 0.16em;
    text-transform: uppercase;
    color: var(--gold);
    margin: 0 0 18px;
}

h1 {
    font-family: 'Fraunces', serif;
    font-weight: 600;
    font-size: 30px;
    line-height: 1.1;
    color: var(--ink);
    margin: 0 0 6px;
    text-shadow: 0 1px 0 rgba(255,255,255,0.5);
}

.role {
    font-family: 'Fraunces', serif;
    font-style: italic;
    font-weight: 500;
    font-size: 15px;
    color: var(--teal);
    margin: 0 0 20px;
}

.bio {
    font-size: 14.5px;
    line-height: 1.65;
    color: var(--ink-soft);
    margin: 0 0 26px;
}

.divider {
    border: none;
    border-top: 1px dashed var(--line);
    margin: 0 0 22px;
}

.meta-row {
    display: flex;
    justify-content: space-between;
    font-family: 'IBM Plex Mono', monospace;
    font-size: 11px;
    color: var(--ink-soft);
    margin-bottom: 24px;
}

.meta-row span strong {
    display: block;
    color: var(--ink);
    font-size: 12.5px;
    margin-top: 3px;
}

.socials {
    display: flex;
    gap: 10px;
}

.socials a {
    flex: 1;
    display: flex;
    align-items: center;
    justify-content: center;
    gap: 7px;
    padding: 11px 8px;
}

```

```

border: 1px solid var(--line);
border-radius: 8px;
background: rgba(255,255,255,0.35);
color: var(--ink);
text-decoration: none;
font-family: 'IBM Plex Mono', monospace;
font-size: 11.5px;
letter-spacing: 0.02em;
transition: transform 0.18s ease, background 0.18s ease, border-color 0.18s ease;
}

.socials a svg { width: 14px; height: 14px; flex-shrink: 0; }

.socials a:hover,
.socials a:focus-visible {
  background: var(--teal);
  color: var(--paper);
  border-color: var(--teal);
  transform: translateY(-2px);
}

.socials a:focus-visible {
  outline: 2px solid var(--gold);
  outline-offset: 2px;
}

@media (max-width: 420px) {
  .card { width: 100%; padding: 40px 24px 26px; }
  .socials { flex-wrap: wrap; }
  .socials a { min-width: 45%; }
}

@media (prefers-reduced-motion: reduce) {
  .socials a { transition: none; }
}
</style>
</head>
<body>

<div class="card">
  <div class="stamp">G</div>

  <p class="eyebrow">My Portfolio</p>

  <h1>Govindamma</h1>
  <p class="role">Software Developer</p>

  <p class="bio">
    I am a btech student in anits college pursuing 3rd year in cse.
    I want to become a software developer.
  </p>

  <hr class="divider">

  <div class="meta-row">
    <span>Based in
      <strong>Vizag, IN</strong>
    </span>
    <span>Available
      <strong>Oct 2026</strong>
    </span>
  </div>

  <div class="socials">
    <a href="mailto:payalagovindamma@gmail.com">
      <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="1.6"><rect x="3" y="5"
width="18" height="14" rx="2"/><path d="M3 7l9 6 9-6"/></svg>
      Email
    </a>
    <a href="https://linkedin.com/in/payalagovindamma" target="_blank" rel="noopener">
      <svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="1.6"><rect x="3" y="3"
width="18" height="18" rx="2"/><line x1="8" y1="10" x2="8" y2="16"/><circle cx="8" cy="7" r="0.6"
fill="currentColor"/><path d="M12 16v-3.5c0-1.5 1-2.5 2.3-2.5S16 11 16 12.5V16"/></svg>
      LinkedIn
    </a>
  </div>
</div>

</body>
</html>

```

7. CODE EXPLANATION

Code	Explanation
<code>:root { --var: value; }</code>	Defines reusable CSS custom properties (variables) for colours and sizing.
<code>var(--name)</code>	Retrieves the value of a CSS custom property wherever it is used.
<code>display: flex</code>	Enables flexbox layout to align and centre child elements.
<code>radial-gradient()</code>	Creates a circular colour gradient used for the background lighting effect.
<code>repeating-linear-gradient()</code>	Creates a repeating pattern, used here to add a subtle grain texture.
<code>box-shadow (layered)</code>	Applies multiple shadow layers to give the card a raised, floating appearance.
<code>border-radius</code>	Rounds the corners of the card, stamp, and social buttons.
<code>::before / ::after</code>	Pseudo-elements used to add decorative content without extra HTML tags.
<code>position: absolute / fixed</code>	Positions the stamp badge and background texture relative to their containers.
<code>transform: rotate() / translateY()</code>	Rotates the stamp badge and lifts social buttons slightly on hover.
<code>:hover</code>	Applies styling changes when the pointer is over a social link.
<code>:focus-visible</code>	Applies a visible outline and matching hover styles for keyboard navigation.
<code>@media (max-width: ...)</code>	Applies alternate styles when the viewport is narrower than the given width.
<code>@media (prefers-reduced-motion: reduce)</code>	Disables transition animations for users who prefer reduced motion.
<code>@import via <link> (Google Fonts)</code>	Loads the Fraunces, Inter, and IBM Plex Mono fonts used across the card.
<code><svg></code>	Embeds vector icons directly for the Email and LinkedIn buttons.

8. TEST CASES AND EXECUTION DATA

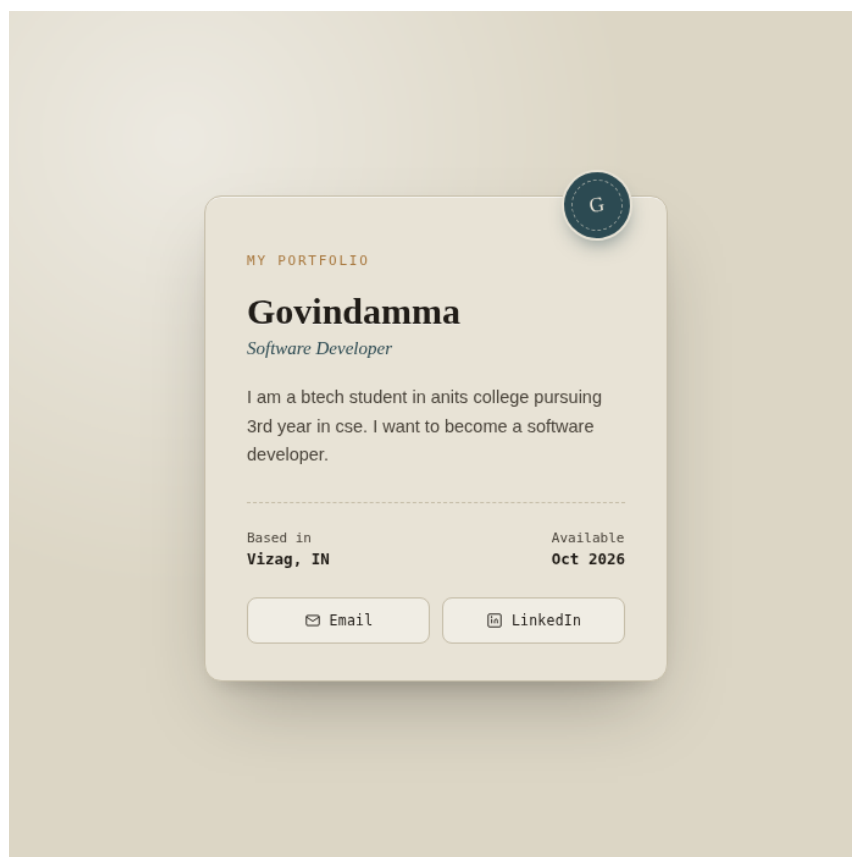
Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Open index.html in a browser	Profile card should render, centred on the page	Card displayed correctly	Pass
TC-02	Check card content	Name, role, bio, location and availability should display	All fields displayed correctly	Pass
TC-03	Hover over the Email button	Background should turn teal and the button should lift slightly	Hover effect applied correctly	Pass
TC-04	Hover over the LinkedIn button	Background should turn teal and the button should lift slightly	Hover effect applied correctly	Pass
TC-05	Tab to a social link using the keyboard	A visible gold focus outline should appear	Focus outline displayed correctly	Pass
TC-06	Click the Email button	Default mail client should open with the address pre-filled	Mail client opened correctly	Pass

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-07	Click the LinkedIn button	LinkedIn profile should open in a new browser tab	New tab opened correctly	Pass
TC-08	Resize the browser window below 420px width	Card should expand to full width and buttons should wrap	Responsive layout applied correctly	Pass
TC-09	Enable reduced motion in OS settings and hover a button	Hover transition should be disabled	Transition removed as expected	Pass
TC-10	Check the circular stamp badge	Badge should display the letter "G" with a dashed inner ring	Stamp displayed correctly	Pass

9. OUTPUT

Output: Profile Card Rendered in Browser

On opening index.html in a browser, a centred profile card is displayed on a warm paper-toned background. The card shows a rotated circular "G" stamp, the eyebrow label "MY PORTFOLIO", the heading "Govindamma", the italic role "Software Developer", a short bio, a dashed divider, a location/availability row, and two social buttons for Email and LinkedIn.



Screenshot: index.html rendered in the browser, showing the styled profile card with stamp, bio, meta row, and social buttons.

10. OBSERVATION / INTERVIEW QUESTIONS

Q1. What are CSS custom properties (variables) and why are they useful?

Answer: CSS custom properties are values defined using a `--name` syntax, typically inside `:root`, and reused throughout a stylesheet with `var()`. They make it easy to maintain a consistent colour palette and update the design from one central place.

Q2. How does Flexbox help centre the profile card on the page?

Answer: Setting `display: flex` along with `align-items: center` and `justify-content: center` on the body element centres its only child, the `.card` div, both vertically and horizontally regardless of viewport size.

Q3. What is the difference between `:hover` and `:focus-visible`?

Answer: The `:hover` pseudo-class applies styles when the pointer is over an element, while `:focus-visible` applies styles only when an element receives keyboard focus, which improves accessibility for keyboard users without adding an outline on every mouse click.

Q4. What is the purpose of the media query for `prefers-reduced-motion`?

Answer: It detects whether the user has requested reduced motion at the operating-system level and, if so, removes the hover transition so the interface does not animate for users sensitive to motion.

Q5. How does the responsive media query change the layout on small screens?

Answer: When the viewport width is 420px or less, the `.card` width switches from a fixed value to 100%, and the `.socials` buttons are allowed to wrap onto a new line with a minimum width, so the card remains usable on mobile screens.

Q6. Why are Google Fonts imported using `<link>` tags instead of `@import` in CSS?

Answer: Using `<link>` tags in the document head allows the browser to start downloading the font files earlier and in parallel with other resources, which is generally faster than an `@import` placed inside a CSS file.

Q7. What visual technique is used to create the card's raised appearance?

Answer: Multiple `box-shadow` values are layered together: one inset highlight along the top edge and two outer shadows at different blur and spread distances, which together simulate soft, realistic depth beneath the card.

11. RESULT

Thus, a webpage was successfully designed using HTML5 and CSS3, combining semantic structure with custom fonts, CSS variables, flexbox layout, layered shadows, gradients, hover/focus interactivity, and responsive media queries. The profile card rendered correctly in the browser and all test cases relating to appearance, interactivity, accessibility, and responsiveness were verified successfully.